

Errata to “Computing μ -Bases of Univariate Polynomial Matrices Using Polynomial Matrix Factorization”

The article [1] contains a few typographical errors, which are corrected as follows.

1. **Page 1193, proof of Proposition 3.1.** Replace P_{1k} with P_{1l} in the notation for the column-interchange matrix.
2. **Page 1193, proof of Proposition 3.1, polynomial division.** Replace $\deg(r_i) < \deg(f_k)$ with $\deg(r_i) < \deg(f_1)$.
3. **Page 1193, proof of Proposition 3.1, second displayed matrix multiplication.** Replace the transformation matrix multiplying $(f_1, \dots, f_n)$ with

$$\begin{pmatrix} 1 & -q_2 & \cdots & -q_n \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

4. **Page 1194, paragraph on degree reduction.** Replace “Proposition 2.2” with “Proposition 2.5”.
5. **Page 1197, proof of Proposition 4.1.** For $i = 1, 2$, replace $(n - i) \times 1$ with $(m - i) \times 1$ in the dimension of $c_i(s)$, and replace $(n - i) \times (n - i)$ with $(m - i) \times (n - i)$ in the dimension of $M_i(s)$.
6. **Page 1198, Example 4.2, item 1).** Replace $M(s) = R(s)Q(s)$ with $M(s)Q(s) = R(s)$.
7. **Page 1201, Section 6.1, first paragraph.** Replace “SG and CH” with “SG and HC”.
8. **Page 1203, Section 6.2, sentence describing Figure 4.** Replace $n = 5, 7, 9, 11$ with $n = 7, 9, 11, 13$.
9. **Page 1206, reference [6].** Replace “Computer Aided Geometric Design” with “Journal of Symbolic Computation”.
10. **Page 1206, reference [15].** Replace “2003” with “2002”.
11. **Page 1206, reference [18].** Replace “Using Algebraic Geomety” with “Using Algebraic Geometry”.

References

- [1] B. Huang, F. Chen, Computing μ -bases of univariate polynomial matrices using polynomial matrix factorization, *Journal of Systems Science and Complexity* **34** (2021), 1189–1206. doi:10.1007/s11424-020-9314-6.